1. Introduction

1.1 Purpose

The BAS aims to automate the operations of a bookshop, facilitating seamless interaction between shopkeepers, customers, and vendors. It provides functionalities for managing inventory, generating bills, handling customer feedback, and managing supply chain operations.

1.2 Document Conventions

- BAS: Book Automation Software
- GUI: Graphical User Interface
- CRUD: Create, Read, Update, Delete

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- Developers: Focus on system features and functional requirements.
- Project Managers: Review user classes, design constraints, and assumptions.
- Testers: Analyze functional requirements for test case creation.

1.4 Product Scope

The BAS will streamline inventory management, sales transactions, and vendor interactions, improving operational efficiency. It will also enhance customer satisfaction through feedback and book demand features.

1.5 References

- IEEE SRS Template
- Web Development Resources (HTML, CSS, JavaScript)
- LocalStorage Documentation for Frontend Applications

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2. Overall Description

2.1 Product Perspective

The BAS is a standalone web application integrating shop management, customer interaction, and vendor communication into one system. It ensures inventory updates after sales and vendor supply transactions.

2.2 Product Functions

- **Shopkeeper**: Add, update, or delete books; manage inventory; view sales statistics.
- **Customer**: Query and purchase books; provide feedback and demand unavailable books.
- **Vendor**: Supply books; communicate with shopkeepers.
- **BAS**: Generate bills and update stock automatically.

2.3 User Classes and Characteristics

- **Shopkeeper**: Moderate technical skills; frequent usage.
- **Customer**: General users; occasional usage.
- **Vendor**: Moderate technical skills; periodic interaction.

2.4 Operating Environment

- Platforms: Modern web browsers (Chrome, Firefox, Safari).
- Hardware: PC or mobile with internet access.

2.5 Design and Implementation Constraints

- Use of localStorage for persistence.
- Limited to single-shop operations.

2.6 User Documentation

Help guides and FAQ sections for shopkeepers and customers.

2.7 Assumptions and Dependencies

- Assumes internet connectivity.
- Requires users to operate in supported browsers.

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3. External Interface Requirements

3.1 User Interfaces

- Interactive forms for book management.
- Search and filter options for books.

3.2 Hardware Interfaces

Basic input devices (keyboard, mouse).

3.3 Software Interfaces

- Browser localStorage for data persistence.
- Use of jsPDF for bill generation.

3.4 Communications Interfaces

Direct links for feedback and messaging between users.

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4. System Features

4.1 Feature 1: Book Management

1. **Description**: CRUD operations on books.
2. **Priority**: High
3. **Functional Requirements**:
    - Add/update books with price, quantity, and image.
    - Remove outdated books.
    - Update stock after transactions.

4.2 Feature 2: Sales Transactions

1. **Description**: Generate bills after purchase.
2. **Priority**: High
3. **Functional Requirements**:
    - Automatic stock deduction.
    - Generate and download sales invoices.

4.3 Feature 3: Feedback and Requests

1. **Description**: Handle customer feedback and book demands.
2. **Priority**: Medium
3. **Functional Requirements**:
    - Submit feedback.
    - View and manage requests.

4.4 Feature 4: Vendor Communication

1. **Description**: Direct messaging and inventory supply.
2. **Priority**: Medium

3. **Functional Requirements**:
    - Update stock from vendor supplies.
    - Message vendors about orders or stock issues.

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5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Quick response time for CRUD operations.
- Efficient data retrieval for large inventories.

5.2 Safety Requirements

- Prevent unauthorized inventory updates.

5.3 Security Requirements

- Authentication for shopkeeper and vendor portals.

5.4 Software Quality Attributes

- **Adaptability**: Mobile and desktop compatibility.
- **Reliability**: Consistent stock updates.

5.5 Other Requirements

- Extendable for multi-shop operations.

Discussion/Conclusion

Function of Requirement Analysis

To ensure a clear understanding of system objectives, user roles, and functional workflows.

Steps in Requirement Analysis

1. Identify stakeholders.
2. Elicit requirements through interviews and brainstorming.
3. Document and validate requirements.

Significance in SDLC

Requirement analysis defines project scope, minimizes risks, and lays the foundation for design and development.